MIL-I-23298A(OS)
5 March 1980
Superseding
MIL-I-23298(OS)
28 January 1963
and Amendment 1
25 August 1971

MILITARY SPECIFICATION
IGNITERS, ROCKET MOTOR, LOADED MK 181 MOD 0 AND MK 200 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the MK 181 MOD 0 and the MK 200 MOD 0 loaded igniters hereinafter referred to as the igniters.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-22450	Booster, Guided Missile, Mark 12 MOD 0 (Loaded Assembly)
MIL-B-23297	Booster, Guided Missile, Mark 12 MOD 1 (Loaded Assembly)
MIL-P-46994	Pellets, Boron Potassium Nitrate
MIL-P-81351	Peroxide, Methyl Ethyl Ketone

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1167	Ammunition Data Card

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

OS11513	Ignition Element, Electric Mark 9 MOD 0

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

LD487619	Igniter, Booster, Guided Missile Mark 181 MOD 0 (For Loading and Service Use)
LD487649	Igniter, Rocket Motor, Mark 200 MOD 0, Loading and Service
1636671	Pressure Squib
1636673	Booster, Guided Missile, Mark 12 MOD 0 Ballistic Data
1636730	Booster, Guided Missile, Mark 12 MOD 1 Ballistic Data
1738546	Igniter Static Test Arrangement (MK 181)
1738575	Igniter Static Test Arrangement (MK 200)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract (see 6.2), first article samples of the igniters are required and shall be manufactured using the methods and procedures proposed for production. The first article shall be tested as specified in Section 4 for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing igniters that comply with the technical requirements of the contract (see 4.5).

3.2 General requirements. The materials, dimensions and performance of the igniters MK 200 MOD 0, and 181 MOD 0, respectively, shall be in accordance with the requirements of this specification and all documents listed on LD487649 and LD487619, respectively.

3.3 Ballistic requirements. The igniters shall meet the ballistic requirements of 3.3.1 or 3.3.2.

3.3.1 Motor functioning.

a. The MK 200 MOD 0 igniter shall be capable of igniting the MK 12 MOD 1 rocket motor so as to meet the ballistic requirements specified in MIL-B-23297 and Dwg 1636673.

b. The MK 181 MOD 0 igniter shall be capable of igniting the MK 12 MOD 0 rocket motor so as to meet the ballistic requirements specified in MIL-B-22450 and Dwg 1636730.

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3.3.2 Igniter static firing.

a. The MK 200 igniter shall meet the requirements of Table I when static fired in an igniter static test arrangement in accordance with Dwg 1738575.

b. The MK 181 igniter shall meet the requirmenets of Table II when static fired in an igniter static test arrangement in accordance with Dwg 1738546.

3.4 Continuity. The circuit resistance of the igniters shall be as specified in Table III

TABLE I. MK 200 igniter ballistics.[1]

Requirement	Limits at -12°C		Limits at 49°C	
	Lower	Upper	Lower	Upper
Maximum Pressure (PSI)	245	345	255	355
Ignition Delay Time (seconds)	---	0.095	---	0.072
Time to Maximum Pressure (seconds)	---	0.112	---	0.106

[1]See 6.7 for definitions

TABLE II. MK 181 igniter ballistics.[1]

Requirement	Limits at -12°C		Limits at 49°C	
	Lower	Upper	Lower	Upper
Maximum Pressure (PSI)	275	365	275	365
Ignition Delay Time (seconds)	---	0.070	---	0.070
Time to Maximum Pressure (seconds)	---	0.130	---	0.130

[1]See 6.7 for definitions

TABLE III. Circuit resistance.

Igniter	Resistance, ohms @ 25°C	
	Safe	Armed
MK 200	0.03 ± 0.02	1.05 ± 0.18
MK 181	----	0.18 ± 0.05

3.5 Igniter Charge.

3.5.1 Igniter pellets. The igniter charge shall be pellets in accordance with type V, Class 1 of MIL-P-46994 with the following exceptions:

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a. The crush strength of the pellets shall be 18,500 grams (g), minimum, radially and 18,000 g, minimum, axially.

b. The organic peroxide catalyst (60% methylethyl ketone peroxide in dimethyl phthalate) shall be in accordance with MIL-P-81351.

3.5.2 Pellet lots. Lots of pellets defined per and meeting the requirements of MIL-P-46994 may be combined and blended to form larger lots to allow uniformity in large igniter lots.

3.5.3 Reuse of pellets. Fleet returned pellets may be reused provided the pellets meet the requirements of 3.5.1 and 3.5.2.

3.6 Ignition source.

3.6.1 Pressure squib. The ignition source for the MK 200 igniter shall be a pressure squib meeting the requirements of Dwg 1636671.

3.6.2 Ignition element. The ignition source for the MK 181 igniter shall be the MK 9 MOD 0 electric ignition element in accordance with OS11513.

3.7 Workmanship. The requirements of workmanship shall be as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part of the assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract,

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the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5)
b. Quality conformance inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspection shall be performed under the following conditions:

a. Temperature: Room ambient 18° to 35°C (65° to 95°F)
b. Humidity: Room ambient but no greater than 95 percent relative

4.4 Certificate of compliance. When specified in the contract (see 6.2), the contractor shall furnish a certificate of compliance stating that the ignition source (pressure squib or electric ignition element) is in conformance with this specification and Dwg 1636671 or OS11513, as applicable.

4.5 First article inspection. Unless otherwise specified in the contract (see 6.2) a first article sample of 20 igniters, manufactured in accordance with 3.1, shall be submitted to an activity designated by the procuring activity (see 6.2) for first article inspection. The first article samples shall have passed the quality conformance inspection of 4.6 with the exception of the static firing test. The ignition source shall have met the requirements of this specification and Dwg 1636671 or OS11513, as applicable. The first article samples shall be subjected to the continuity test of 4.7.3, then test fired in accordance with 4.7.4. Acceptance of the first article sample shall be based on no defects in the sample. Further production of igniters by the contractor prior to approval of the preproduction sample shall be at the contractor's risk. First article samples shall not be included as a part of the quantity specified for delivery in the contract (see 6.2). When specified in the contract a first article inspection report shall be prepared and submitted to the procuring activity.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests of 4.7 conducted on samples selected as specified in 4.6.2. The ignition source and charge pellet tests of 4.7.1 and 4.7.2 shall be performed on the lots in question prior to their installation into igniters. The contractor is responsible for assuring that all igniters offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number. Ballistic samples shall not be included in the quantity specified for delivery on the contract (see 6.2).

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4.6.1 Test reports. When specified in the contract (see 6.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information:

a. Static firing performance test results (see 4.7.4)
b. Ammunition data cards per MIL-STD-1167.

4.6.2 Sampling for quality conformance inspection.

4.6.2.1 Lot. For purposes of identification, inspection, and surveillance, a lot of igniters shall consist of those igniters produced by one manufacturer at one plant, on one production line, containing one lot of squibs or ignition elements, and containing pellets from only one lot or one blended lot.

4.6.2.2 Sampling for ignition source inspection.

a. Sampling for pressure squib no-fire test shall be in accordance with inspection level II of MIL-STD-105. Sampling for pressure squib all-fire and pressure output tests shall be in accordance with inspection level S-4 of MIL-STD-105.
b. Sampling for MK 9 MOD 0 electric ignition element tests and examinations shall be as specified in OS11513.

4.6.2.3 Sampling for charge pellet inspection. Sampling for charge pellet inspection shall be in accordance with the following:

a. Igniter pellet lots, composed of pellets removed from Fleet returned igniters and blended to form a homogeneous lot, shall be sampled in accordance with MIL-P-46994.
b. Fleet returned loaded igniters shall be sampled in accordance with inspection level I of MIL-STD-105. The sample unit shall be the igniter assembly. The pellets shall be removed from the sample igniters and inspected in accordance with 4.7.2. The original igniter lot identification and integrity shall be maintained.

4.6.2.4 Sampling for continuity test. Each igniter to be submitted for acceptance on contract shall be subjected to the continuity test of 4.7.3.

4.6.2.5 Sampling for ballistics test. Sampling for igniter ballistics inspection shall be in accordance with inspection level S-4 of MIL-STD-105.

4.6.2.6 Sampling for examination. Sampling for examination for workmanship and drawing characteristics not covered in the above sampling requirements shall be as follows:

a. Critical - Sample 100% of lots up to 150 igniters. Sample larger lots in accordance with inspection level II of MIL-STD-105.
b. Major and minor - Sample in accordance with level II of MIL-STD-105.

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4.7 Tests and examinations.

4.7.1 Ignition source tests.

4.7.1.1 Pressure squib no-fire test. Each sample squib selected in accordance with 4.6.2.2 shall be subjected to a current of 0.50 + 0.00, -0.05 amperes (A) for 5 minutes, minimum. Any indication of ignition of any squib shall result in rejection of the represented lot.

4.7.1.2 Pressure squib all-fire and pressure output test. Each sample squib selected in accordance with 4.6.2.2 shall be subjected to a current of 2.0 + 0.1,-0.0 A for 50 milliseconds, minimum. The sample shall meet the all-fire and output requirements of Dwg 1636671 with an acceptable quality level (AQL) of 1.0% defective.

4.7.1.3 MK 9 MOD 0 electric ignition element tests. MK 9 MOD 0 electric ignition element tests and acceptance criteria shall be in accordance with OS11513.

4.7.2 Charge pellet tests and examination. Sample pellets, selected in accordance with 4.6.2.3, shall be subjected to the tests and examinations of MIL-P- 46994. Acceptance criteria shall be as specified in MIL-P-46994.

4.7.2.1 Ballistic proofing. One igniter shall be loaded and test fired in accordance with 4.7.4.1 for each lot of pellets to be used in igniter production. This ballistic proofing test is to insure that the pellet lot is capable of meeting specification requirements. Failure to meet the requirements of 3.3.1 shall result in rejection of the pellet lot.

4.7.3 Continuity test. Each igniter shall be subjected to a continuity test at 25°C to verify conformance with 3.4. The test equipment shall be approved by the procuring activity. Any igniter failing the continuity test shall be rejected and removed from the lot.

4.7.4 Ballistic test. Each sample igniter selected in accordance with 4.6.2.5 shall be subjected to test of 4.7.4.1 or the alternate test of 4.7.4.2. The samples for a given lot may be subjected to any combination of static and motor functioning tests to allow conformance with delivery schedules. Acceptance for ballistics shall be based on conformance with the requirements of 3.3.1 and 3.3.2 with an AQL of 1.5% defective. In case of a dispute the static test of 4.7.4.1 shall be determinative.

4.7.4.1 Igniter static test. The sample MK 200 or MK 181 igniters shall be test fired in an igniter static test arrangement in accordance with Dwg 1738575 or 1738546, respectively. One half of the sample igniters shall be conditioned at 49°C and one half at 10°F for a minimum of eight hours and test fired within 15 minutes after removal from the conditioning chamber.

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4.7.4.2 Alternate motor functioning test. The sample MK 200 or MK 181 igniters shall be installed in a MK 12 MOD 1 or MK 12 MOD 0 rocket motor, as applicable, and conditioned and fired with the motor at 25 °C per MIL-B-23297 or MIL-B-22450. The motor ballistics shall then be reviewed for igniter related anomalies. Motor failures which are attributed to igniter performance (i.e., ignition delay time) shall be considered a failure of igniter ballistic test.

4.7.5 Examination. Sample igniters and components selected in accordance with 4.6.2.6 shall be examined to verify conformance to all specification and drawing requirements. The AQL's shall be 0.10% defective for critical characteristics, 1.0% defective for majors, and 2.5% defective for minors. Known defective igniters shall not be offered for acceptance on contract.

4.8 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be level A.

5.1.1 Level A. Preservation, packaging and packing shall be as specified in the contract (see 6.2). As a minimum containers shall be in accordance with the requirements of 49 CFR 171-178 for this type of item and shall conform to Uniform Freight Classification, National Motor Freight Classification, or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-178.

6. NOTES

6.1 Intended use. The MK 200 MOD 0 igniter is intended to be used in MK 12 MOD 1 rocket motor. The MK 181 MOD 0 igniter is intended to be used in the MK 12 MOD 0 rocket motor. The MK 12 MODs 0 and 1 rocket motors are used as the booster motor for the Terrier and Standard Missile (extended range) systems.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification
b. Quantity required for delivery (does not include first article samples and lot acceptance ballistics samples, see 4.5 and 4.6)
c. Whether MK 200 or MK 181 igniters are required
d. When a first article sample is not required (see 3.1, 4.5, 6.4 and 6.3)
e. Inspection conditions if other than as specified (see 4.3)
f. Assigned activity for first article inspection (see 4.3)
g. Packaging requirements if other than level A (see 5.1)
h. Specify packaging requirements for level A, if required (see 5.1.1)
i. Any special markings required (see 5.2)
j. Safety precautions (see 6.5)
k. That the requirements of MIL-Q-9858 apply (see 6.6)

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6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

PARAGRAPH	DATA REQUIREMENTS	APPLICABLE DID
4.4	Certificate of Compliance	DI-E-2121
4.5	First Article Inspection Report	DI-T-4902
4.6.1	Acceptance Test Report	DI-T-3721
4.6.1	Ammunition Data Card	DI-E-2001

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article sample. When a first article sample is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article sample.

6.4 First article sample waiver. First article samples submitted and approved on a recent contract (a contract remaining in production or a contract out of production for less than one year) may be accepted by the procuring activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2.1(d)), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract.

6.5 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DAR 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.6 Quality program requirements. Igniters in accordance with this specification are complex-critical items as defined in DAR 14-101.5 and 14-304, and a quality program per MIL-Q-9858 should therefore be specified in the contract.

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6.7 Definitions. The following definitions apply to igniter ballistics (see FIGURE 1):

a. Maximum Pressure (Pmax). The maximum pressure obtained on the pressure - time curve.

b. Ten percent Maximum Pressure (10% Pmax). Ten percent of maximum pressure as defined in a.

c. Ignition Delay Time (td). The time interval between the application of ignition current to the igniter and ten percent maximum pressure.

d. Time to Maximum Pressure (t_{Pmax}). The time interval from ten percent maximum pressure on the rise slope to maximum pressure.

6.8 Changes from previous issue. Due to the comprehensive nature of this revision the margins of this specification have not been marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. Bidders and contractors are advised to evaluate the requirements of this document based on the entire content irrespective of the relationship to the last previous issue.

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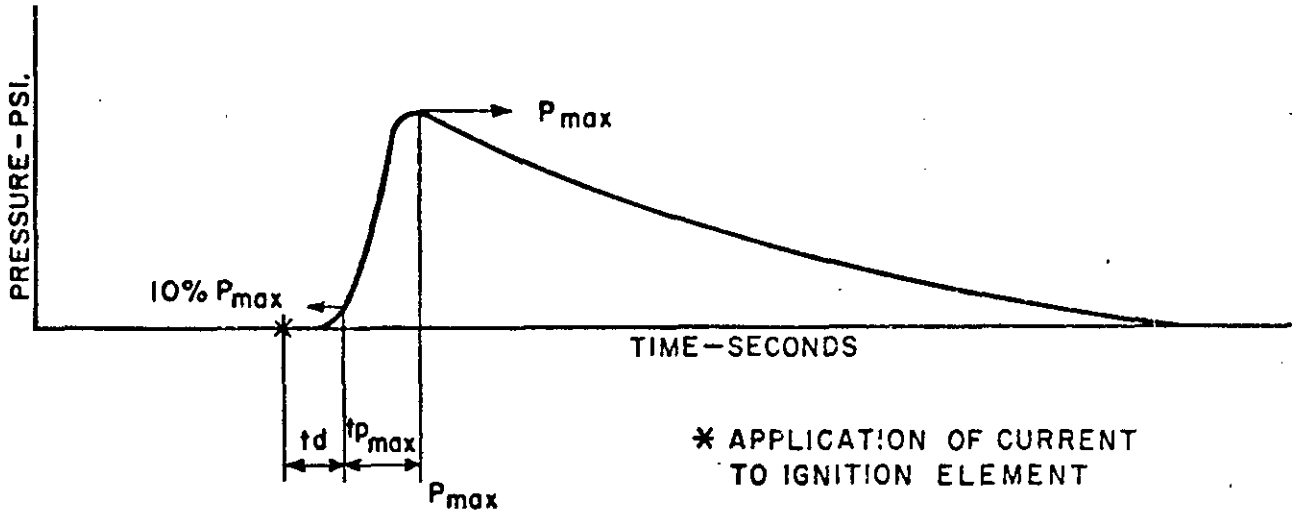


FIGURE 1. Representative igniter static firing pressure - time curve.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE "IGNITERS, ROCKET MOTOR, LOADED MK 181 MOD 0 MIL-I-23298A(OS) AND MK 200 MOD 0"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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